Grade: Check Plus

2. Expected Value and Options

x	f(x)	μ	σ^2	v	g(v) = P(V=v)
20	.05	1.00	1.46	0	0
21	.06	1.26	1.17	0	0
22	.07	1.54	.08	0	0
23	.08	1.84	.46	0	0
24	.09	2.16	.18	0	0
25	.12	3.00	.02	0	.47
26	.15	3.90	.05	1	.15
27	.13	3.51	.64	2	.13
28	.10	2.80	.67	3	.10
29	.10	2.90	1.29	4	.10
30	.05	1.50	1.05	5	.05
	1.00	**25.41**	**7.5**		**1.00**

✓ Q1

1) μ = 25.41 **μ = x · f(x)**

σ = 2.74 $\sigma^2 = \Sigma(x - \mu)^2 \cdot f(x)$

$\sigma = \sqrt{\sigma^2}$

μ = $25.41 σ = $2.74

✓ Q2a

2a) μ − σ = 22.67 **22.67 ≤ X ≤ 28.15**

μ + σ = 28.15 23 ≤ X ≤ 28

23, 24, 25, 26, 27, 28

.08 + .09 + .12 + .15 + .13 + .10 = .67

Probability = 67% within 1 standard deviation

2b) μ − 2σ = 19.93 **all values are within this range**

μ + 2σ = 30.89 ✓ Q2b

19.93 ≤ X ≤ 30.89

20 ≤ X ≤ 30

Probability = 100% within 2 standard deviations

3)

Value (V)	g(v)
0	.47
1	.15
2	.13
3	.10
4	.10
5	.05

✓ Q3

✓ Q4

4) Probability V > 0

.15 + .13 + .10 + .10 + .05 = .53

Prob = 53%

5)

(v)	g(v)	μ	σ^2
0	.47	0	.87
1	.15	.15	.02
2	.13	.26	.65
3	.10	.30	.27
4	.10	.40	.70
5	.05	.25	.06
		1.36	**2.57**

✓ Q5

μ = 1.36

σ = √2.57 = 1.60

μ = $1.36 σ = $1.60